MILESTONE 2

Dataset
Source: Philippine Statistics Authority (PSA)
Coverage: Eight selected regions in the Philippines
(NCR, CAR, Region I,II, III, IV-A, IV-B, V)
Year: 2024
Initial Size: 928 rows and 65 columns
Information: Region, cause of death, age-group mortality counts,
and sex-specific mortality counts

1. Data Loading and Preprocessing

Each row in this dataset represents a specific area and a single cause of death.Instead of showing just one total number for deaths, the columns break the numbers down by age group and sex (for example: males aged 0–4, females aged 0–4, or males aged 5–9). This detailed layout makes it easy to see exactly how different groups of people are affected and helps you spot specific health trends.

The preprocessing process will include the following steps:

1. Data Loading: The CSV file will be loaded using Python's built-in CSV module, when NumPy will be used to convert numerical features into arrays.
2. Data Cleaning: Correct inconsistent region names and cause-of-death labels, checking numerical values for invalid entries, and ensure that mortality counts are properly formatted. Zero mortality counts will be retained since they represent valid observations.
3. Category Verification: Examine the cause-of-death classifications to identify overlapping parent and subcategories.
4. Feature Selection: Selecting male and female mortality counts across the individual age groups as numerical features for the analysis.
5. Dataset Preparation for Modeling: K-Means will use the selected mortality distribution features for clustering. K Nearest Neighbor and Perceptron will use the prepared features and corresponding High/Low mortality labels for supervised classification. The labeled dataset will be divided into training and testing sets for model evaluation.

2. Train/Test Split, Cross-Validation Plan, and Evaluation Metrics

The dataset was divided into 80% training data (737 observations) and 20% testing data (183 observations) from the 920 rows of cleaned data. The training set was used to help the KNN model learn patterns and classify mortality observations, while the testing was used to evaluate how well the model classifies data it has not trained on. A stratified split was applied to maintain a similar proportion of High and Low Mortality Impact categories in both sets.

Evaluation Metrics

For K-Means:

Because unsupervised clustering operates without ground-truth labels, traditional cross-validation or train/test splitting is non-applicable. Instead, internal cluster validation via Silhouette Analysis was implemented from scratch to evaluate cluster cohesion and separation.

For K Nearest Neighbor:

Since KNN is a supervised classification algorithm, an 80:20 train-test split was used to evaluate its performance. Macro F1-Score was used to measure how well the model classified both mortality categories, while the Confusion Matrix showed the number of correct and incorrect predictions. Different K values (3,5,7,and 9) were tested to identify the best performing configuration.

For Perceptron

3. Initial Results

With K-Means:

The model reached convergence on iteration 19. Assessment with k = 6 yielded a Silhouette Score of 0.7192, highlighting key regional epidemiological patterns and demographic trends.

Key Analytical Insights:

1. Infant Mortality Isolation (Cluster 2): The model successfully identified a distinct cluster characterized almost exclusively by under-1 and early childhood age brackets, proving its capacity to separate distinct life-stage vulnerabilities.
2. Urban Health Burden (Cluster 5): Concentrated exclusively within the National Capital Region (NCR), capturing high burdens of metabolic, hypertensive, and stroke-related deaths.

3. Senior Longevity Gap: Across high-volume adult clusters, *85 and Over Female* consistently ranks as a top feature, reflecting female longevity demographics in national mortality data.

Cluster	Record Count	Dominant Demographic Features	Epidemiological / Geographic Focus
0	5	65–69 Male, 85+ Female, 60–64 Male	High-volume Circulatory System/Heart Disease in NCR, Region III & IV-A
1	760	85+ Female, 65–69 Male, 60–64 Male	Baseline national mortality across standard regional causes
2	7	Under 1 Male, Under 1 Female, 1–4 Female	Infant & Perinatal Mortality (Congenital Anomalies, Birth Trauma)
3	103	65–69 Male, 70–74 Male, 60–64 Male	Moderate chronic senior mortality clusters across key provinces
4	15	65–69 Male, 70–74 Male, 85+ Female	High-incidence Neoplasms (Cancers) & Endocrine Diseases
5	30	85+ Female, 70–74 Male, 65–69 Male	NCR-Specific Urban Lifestyle Diseases (Diabetes, Cerebrovascular, Hypertension)

With K Nearest Neighbor:

The model achieved its highest Macro F1-Score of 0.9836 using K = 5, correctly classifying 180 out of 183 testing observations. This configuration performed better than the other tested K values.

Key Analytical Insights:

1. Best Configuration (K=5): Achieved the highest Macro F1-Score and the fewest incorrect predictions among the four configurations.
2. High Classification Performance: The model correctly classified all 91 Low Mortality Impact observations across all tested K values.
3. High Mortality Misclassification: Most classification errors involved High Mortality Impact observations being predicted as Low.

4. Effect of K Values: Performance decreased as K increased by beyond 5, suggesting that a larger neighborhood did not improve classification for this dataset.

K Value	Macro F1-Score	Correct Predictions	Incorrect Predictions
3	0.9781	179	4
5	0.9836	180	3
7	0.9672	177	6
9	0.9562	175	8

Basically, we wanted to know whether KNN could use regions and cause-of-death information to classify mortality observations, and the Macro F1-Score helped in measuring how accurate it was.

With Perceptron